

Sensitivity-related illness: the escalating pandemic of allergy, food intolerance and chemical sensitivity.

The prevalence of allergic-related diseases, food intolerance, and chemical sensitivities in both the pediatric and adult population has increased dramatically over the last two decades, with escalating rates of associated morbidity. Conditions of acquired allergy, food intolerance and chemical hypersensitivity are frequently the direct sequelae of a toxicant induced loss of tolerance (TILT) in response to a significant initiating toxic exposure. Following the primary toxicant insult, the individuals become sensitive to low levels of diverse and unrelated triggers in their environment such as commonly encountered chemical, inhalant or food antigens. Among sensitized individuals, exposure to assorted inciting stimuli may precipitate diverse clinical and/or immune sequelae as may be evidenced by clinical symptoms as well as varied lymphocyte, antibody, or cytokine responses in some cases. Recently recognized as a mechanism of disease development, TILT and resultant sensitivity-related illness (SRI) may involve various organ systems and evoke wide-ranging physical or neuropsychological manifestations. With escalating rates of toxicant exposure and bioaccumulation in the population-at-large, an increasing proportion of contemporary illness is the direct result of TILT and ensuing SRI. Avoidance of triggers will preclude symptoms, and desensitization immunotherapy or immune suppression may ameliorate symptomatology in some cases. Resolution of SRI generally occurs on a gradual basis following the elimination of bioaccumulated toxicity and avoidance of further initiating adverse environmental exposures. As has usually been the case throughout medical history whenever new evidence regarding disease mechanisms emerges, resistance to the translation of knowledge abounds. Copyright © 2010 Elsevier B.V. All rights reserved.